Xuezhi Cang, Ph.D.

Email: xuezhicang@gmail.com **Tel:** 8159811358 **Website:** www.xuezhicang.com
Google Scholar: https://scholar.google.com/citations?user=a3OHfGUAAAAJ&hl=en

Research Interests

GIS, RS, Spatial statistics and GeoAI

Education

2021	Ph.D. in Geography. Northern Illinois University, DeKalb, IL. Dissertation: A "Warm" or "Cold" Early Mars: Evidence from Valley Networks
2021	M.S., Computer Science. Northern Illinois University, DeKalb, IL
2014	M.S., Cartography and Geographic Information Science. Capital Normal University, China. Thesis: Monocular Visual Simultaneous Localization and Mapping
2010	B.S., Geography Information System. Nanjing Normal University, China. Thesis: Large watershed extraction method-using Poyang Lake as an example

Professional Experience

Sep, 2024 -Sep, 2025 — **Postdoctoral Research Scholar**, University of Illinois Chicago, IL
Project: Understanding the structure of depressions on Digital Elevation Model (DEM)

- Designed a parameterized data processing workflow using Python and multiprocessing
- Extracted depression hierarchy from DEM data, converted binary depression hierarchy trees into general trees, and extracted trees' hierarchical layers.
- Leveraged vectorized operations to transform the processing time of large 2D data matrices from several days to just minutes.
- Assessed their correlation with environmental factors by machine learning methods

Sep, 2021 -Aug, 2024 — **Postdoctoral Research Scholar** University of Maryland Baltimore County, MD
Project: Mapping Stream from High resolution DEM

- Managed and coordinated terabyte-level spatial datasets
- Maintained the stream extraction scripts and improved the mapping accuracy and speed
- Designed and implemented the flow routing algorithm based on local landscape pattern, land cover data, and elevation
- Designed and built the stream conflation algorithm to aggregate stream to a higher hydrologic unit
- Deployed the program on High Performance Computing (HPC) and developed a HPC task assignment script to optimize the data processing efficiency

May, 2018– Aug, 2018 — **Summer GIS Intern** Illinois State Water Survey, Champaign, IL
Project: Web mapping and data processing by using Python.

- Leveraged automatic mapping library, i.e. arcpy.mapping, to analyze and generate more than 150 climate data trend maps from 1815 to 2017
- Built webpages visualizing pulled historical climate data using ArcGIS Web APIs

Aug, 2014 – May, 2021	**Teaching and Research Assistant**, Northern Illinois University, DeKalb, IL Project: Early mars geomorphology and spatial statistics

- Developed a valley networks (VNs) mapping program in Python
- Applied machine learning to estimate the length of "warm" Mars duration
- Built a novel spatial analysis method (https://github.com/xuezhicang/SPADE)

Lab instructor: GIS, Remote Sensing, Hydrology, and Applied Spatial Statistics

- Designed ArcGIS online and ArcGIS StoryMaps lab tutorials for undergraduate and graduate students and graded assignments
- Designed Google Earth Engine and Earth Engine Apps lab instructions and examples for students (https://xuezhicang.users.earthengine.app/view/nighttimeandpopulation03)

Peer-reviewed Publications

Full presentation list and link: https://scholar.google.com/citations?user=a3OHfGUAAAAJ&hl=en

2019	**Cang, X**., Luo, W., 2019. Noachian Climatic Conditions on Mars Inferred from Valley Network Junction Angles. Earth and Planetary Science Letters. 526, p.115768. (IF = 4.637)
	Luo, W., Howard, A.D., **Cang, X.**, 2019. Comment on "The volume of water required to carve the Martian valley networks: Improved constraints using updated methods." Icarus (IF = 3.565)
2018	**Cang, X.**, Luo, W., 2018. Spatial association detector (SPADE). Int. J. Geogr. Inf. Sci. 32, 2055–2075. (IF = 3.545)
2017	Luo, W., **Cang, X.**, Howard, A.D., 2017. New Martian valley network volume estimate consistent with ancient ocean and warm and wet climate. Nat. Commun. 8, 15766. (IF = 11.878)
2016	Luo, W., Jasiewicz, J., Stepinski, T., Wang, J., Xu, C., **Cang, X.**, 2016. Spatial association between dissection density and environmental factors over the entire conterminous United States. Geophys. Res. Lett. 43, 692–700. (IF = 4.578)
2015	Li, H., Zhang, A., Hu, S., Huang, H., **Cang, X**., Sun, W., 2015. New Remote Sensing Image Subpixel Registration of Spatial and Frequency Domain. J. Chinese Comput. Syst. 36, 591–596. (in Chinese)
2010	**Cang, X**., Tang, G., Zhong, T., Li, R., 2010. Classification of Peaks and Digital Expression of Their Spatial Pattern. J. Nanjing Norm. Univ. (Natural Sci. Ed). 33, 136–140. (in Chinese)
	Li, R., Teng, F., **Cang, X**., Tang, G., Li, F., 2010. Design of XU Xiake Travel Scientific Popularization Electronic Map. Geomatics World 3. (in Chinese)

Technical document

2025	Matthew E Baker, David Saavedra, **Xuezhi Cang**, Labeeb Ahmed, Hydrography Mapping Supporting Modeling and Targeted Conservation: Project Overview and Lessons Learned https://doi.org/10.5066/P1GRAPEX.

Conference Presentations

2025	**Cang, X**., Callaghan, K., Landscape Structure Through Depression Hierarchy Analysis in the Continental United States, in 2025 CSDMS annual meeting
2023	**Cang, X**., Stream Fusion between Consecutive Watersheds, in: 2023 AAG annual Meeting.
2022	**Cang, X.**, Drainage Maturity: an Indicator Measuring the Duration and Intensity of Fluvial Process by Comparing the Flow Direction Configurations of Real World Landscape and of Optimal Channel Network, in: 2022 AGU Fall Meeting.
2020	**Cang, X.**, Luo, W., Hack's law of Mars Valley Networks, in: 2020 AGU Fall Meeting.
2019	**Cang, X.**, Luo, W., Utilizing Valley Network Junction Angle to Estimate the Duration of "Warm" Mars, in: 50th Lunar and Planetary Science Conference (2019).
2017	**Cang, X.**, Luo, W., Frequency Distribution of Junction Angles of Valley Networks on Mars Consistent with an Early Warm Climate, in: Fourth International Conference on Early Mars: Geologic, Hydrologic, and Climatic Evolution and the Implications for Life.
2016	**Cang, X.**, Luo, W., Spatial Association between Valley Density and Environmental Factors over the whole of Conterminous China, in: The 33rd International Geographical Congress.
	Cang, X., Luo, W., An Improved Spatial Association Estimator under the Geographical Detector Model Using Monte Carlo Simulation, in: 2016 AGU Fall Meeting.
2015	**Cang, X.**, Luo, W., Area Measurement Errors in Equal-area Projection, in: 2015 AAG Annual Meeting.
2009	Zhong, T., **Cang, X.**, Li, R., Tang, G., Landform classification based on hillslope units from DEMs, in: 30th Asian Conference on Remote Sensing (ACRS) Proceedings

Book Chapter

2018	Luo, W., Hartmann, J., Wang, F., Pingwen, H., Sysamouth, V., Li, J., **Cang, X.**, 2018. GIS in Comparative-Historical Linguistics Research: Tai Languages, in: Comprehensive Geographic Information Systems. Elsevier, pp. 157–180.

Teaching Experience:

2018 Aug –	**Teaching Assistant (Lab Instructor),** Northern Illinois University, DeKalb, IL
2020 May	Courses: GIS, RS, Hydrology, and Spatial Statistics
2012 summer	**Teaching Assistant,** Capital Normal University, Beijing, China
2013 summer	Courses: summer GIS course, and summer RS course

Honors & Scholarships

2025	Travel Scholarships for 2025 CSDMS Annual Meeting
2020	Dissertation Completion Fellowship ($ 14,000)
2020	Richard E. Dahlberg Memorial Scholarship of Northern Illinois University ($ 300)
2019	Student Scholarship of Illinois GIS Association ($1,000, awarded to one student annually)
2019	Outstanding Graduate Student Award of Northern Illinois University
2018	Richard E. Dahlberg Memorial Scholarship of Northern Illinois University ($ 500)
2017	Great Journeys Graduate Assistantship ($ 16,000)

Service

From 2019	From 2019 Peer-review for the following journals: International Journal of Geographical Information Science, Computational Urban Science, and Journal of Geographical Systems
2014, 2015 and 2017	NIU STEM fest Volunteer